An Eclipse Stride Pattern with Off-Stride Eclipses Sharing Intervals with Long-Term Cycles in a Thirty Thousand Year Eclipse Catalog

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Abstract

A catalog of every solar and lunar eclipse from −13198 to +17189 was computed from NASA JPL's DE441 ephemeris: 145,356 eclipses, about 4.78 per year. A fixed stride rule was applied to the whole sequence: step 23.5 synodic months (694 days) to the next eclipse, or 34.5 synodic months (1,019 days) when no eclipse lies at 23.5. Walks started at both ends of the catalog settle into six forward and six backward lanes that together reach 140,232 eclipses (96.5%).

The 5,124 eclipses that no lane reaches (off-stride eclipses) are all partial solar or penumbral lunar eclipses with $|\gamma|$ between 1.2104 and 1.5853. All lie within 16 eclipses of the start or end of their Saros series.

From every off-stride eclipse, the same 16,377 intervals lead to other eclipses, reaching up to about 4,800 years into the past and the future. For comparison, the same test run from every eclipse among the first or last 15 members of each Saros series (59,498 eclipses) finds 8,268 shared intervals, and from only the first and last members (3,971 eclipses) finds 15,270. The intervals apply in one direction for eclipses near the start of a Saros series (Direction A) and in the opposite direction for those near the end (Direction B).

In time order, the off-stride eclipses form 112 alternating A/B cycles. Their eclipse counts and lengths fall steadily across the record. A further 2,854 on-stride eclipses and 2,993 eclipse near-misses (new or full moons with no eclipse) also match all 16,377 intervals.

The data and code are published so the findings can be recalculated, validated, and further understood.

1. Introduction

This report describes regularities found in a computed 30,388-year eclipse record. It states what was calculated, how, and what the results are, with every parameter needed to reproduce them.

The findings, in the order presented:

1. **On-stride and off-stride eclipses** (Section 3). A 23.5/34.5-synodic-month stride rule divides the eclipse record into eclipses the stride reaches (on-stride) and eclipses it never reaches (off-stride).
2. **The shared interval set** (Section 4). A common set of 16,377 intervals leads from every off-stride eclipse to other eclipses. It applies in one of two directions, according to whether the eclipse is near the start or the end of its Saros series.
3. **Off-stride cycles** (Section 5). In time order, the off-stride eclipses form alternating same-direction cycles, whose size and length change steadily across the record.
4. **Additional matching points** (Section 6). Some on-stride eclipses, and some new and full moons where no eclipse occurs, also match the full interval set.

Section 7 summarizes the findings. The catalog, code, and key data files are available in the accompanying GitHub repository (Section 8). Readers are invited to recalculate the results, test them, and help explain them.

Notation. Years use astronomical numbering, which includes a year 0: year 0 = 1 BCE and −1 = 2 BCE. Dates are written MM/DD/YYYY, with negative years shown in parentheses; for example, 08/28/(13192) is 28 August of astronomical year −13192, i.e. 13193 BCE. Years standing alone in the text and figures carry a minus sign (e.g. −13198). The Julian calendar is used before 1582 October 15 and the Gregorian calendar from that date. Times are Terrestrial (Dynamical) Time, TD. A *Saros series* is the family of eclipses separated by 6,585.32 days (about 18 years 11 days). The *Inex* is the 10,571.95-day interval (358 lunations, about 29 years less 20 days) that links neighboring Saros series (van den Bergh 1955).

2. The Eclipse Catalog

2.1 Computation

- **Positions:** apparent geocentric positions of the Sun and Moon from the JPL DE441 ephemeris (Park et al. 2021), read with the Skyfield library (Rhodes 2019); array calculations use NumPy (Harris et al. 2020).
- **New and full moons:** every one between −13198 and +17189 was located directly from the ephemeris, starting from the mean-lunation formula of Meeus (1998, Ch. 49). Any with the Moon more than 1.75° from the shadow axis was skipped, since no eclipse is possible there (the true limit is about 1.58°).
- **Greatest eclipse:** the instant the shadow axis passes closest to Earth's center (solar), or the Moon passes closest to the center of Earth's shadow (lunar). Gamma (γ) is that closest distance, in Earth radii.
- **Type and magnitude:** from standard shadow geometry (Meeus 1998, Ch. 54). Two constants were calibrated to NASA's published values: a 63.5 km atmospheric enlargement of Earth's shadow, and a lunar radius of $k = 0.272307$ for solar magnitudes.
- **Time:** all eclipse times are TD, taken directly from the ephemeris. ΔT (Espenak & Meeus 2006, based on Morrison & Stephenson 2004, with NASA's lunar-acceleration correction following Chapront et al. 2002) only fills the catalog's ΔT column. It is not used in any calculation in this report.
- **Result:** 145,356 eclipses (72,224 solar, 73,132 lunar). The first eclipse in the catalog is on 06/06/(13198) and the last on 08/25/17189.

2.2 Comparison with other catalogs

Table 1 compares the catalog with three independent sources and summarizes internal consistency checks.

Table 1. Comparison of the catalog with independent catalogs.

Compared with	Range	Result
NASA Five Millennium Catalogs (Espenak & Meeus 2006, 2009)	1001 to 2000	All 4,816 eclipses reproduced, none missing. Mean time difference 0.66 s (solar), 3 s (lunar); γ within 0.00004; 6 type differences
Yuk Tung Liu's eclipse catalogs (Liu n.d.)	−3999 to +4000	Identical counts (19,013 solar, 19,271 lunar). All dates match except one near-zero-magnitude grazing eclipse on each side. γ identical to 4 decimals; solar times within 1 s, lunar within 13 s
Saros-Inex panorama (Quaglia & Tilley n.d.)	−11000 to +15000	61,231 of 61,775 solar eclipses at identical Saros/Inex positions
Internal consistency	Full span	No Saros series missing an interior member; 4.76 to 4.81 eclipses per year in every 2,000-year block; solar/lunar ratio 0.984 to 0.990

2.3 Conventions and limitations

- **Edges of the comparisons:** no external catalog covers years before −3999 or after +4000. There the catalog rests on DE441 alone.
- **Faint grazing eclipses:** near-zero-magnitude eclipses at the very start or end of a Saros series can differ between calculations. For example, NASA lists 78 eclipses in Saros 69; this catalog and Liu's include one more, of magnitude 0.0007.
- **Total vs hybrid:** a few hybrid eclipses are labeled total. Eclipse type is not used by any finding in this report.
- **Master time axis:** cumulative seconds since −15000 January 1, 00:00 TD (Julian calendar before 1582 October 15, Gregorian after; calendar conversion follows Meeus 1998, Ch. 7). Every interval in this report is a difference of cumulative seconds. The first catalog year is partial; it begins on Julian −13198 April 12.
- **Saros numbers:** from −3999 to +4000, numbered by the author to match NASA's Saros numbering wherever it could be verified. Outside that range they are extended by the standard numbering rule.
- **Saros positions:** each eclipse has two Saros positions, counted from the first and from the last member of its series; position 1 is the first or last member itself. Positions in series cut off by the catalog edges are estimated by extending the series' γ trend, and are marked "~".

3. On-Stride and Off-Stride Eclipses

3.1 The stride rule

From an eclipse, step forward 23.5 synodic months. If an eclipse lies within ±1.5 days (129,600 seconds) of that point, it is the next step. If not, step 34.5 synodic months instead. Repeat from each eclipse reached, to the end of the catalog. The same walk is also run backward in time.

23.5 synodic months (about 694 days) is about two eclipse years, so each step moves between a solar and a lunar eclipse. 34.5 synodic months is about 1,019 days. 23.5 and 34.5 synodic months are common eclipse intervals because they nearly coincide with 25.5 and 37.5 draconic months, and in both units the counts are half-integers.

The rule was found through exploratory testing, not derived from theory, and is reported here as found. The same off-stride set results for any stride tolerance from ±1.5 to ±3.5 days, for strides of 693 or 695 days in place of 694 and of 1,018 or 1,020 days in place of 1,019, and for seed windows (Section 3.2) of 694, 1,019, or 1,100 days.

3.2 Lanes

- **Forward lanes:** 6 distinct forward lanes run from the start of the catalog. Their starting eclipses were determined to be true lanes by matching the eclipses the lanes miss to eclipses that match all of the off-stride shared intervals (Section 4). This was needed only at the catalog edges: all walks eventually converge into the same unique lanes, but at the edges there are no earlier or later eclipses from which to tell which starting points are true lanes.
- **Backward lanes:** 6 distinct backward lanes run from the end of the catalog, with their starting eclipses identified in the same way. These 6 forward and 6 backward lanes are the total number of lanes possible: a walk started from a different eclipse at the start or end of the catalog quickly converges with one of them and follows it as a single lane for the rest of the catalog.
- Each lane visits about 15,750 eclipses and runs the full span of the catalog.
- **On-stride eclipses** lie on at least one lane: 140,232 (96.5%).
- **Off-stride eclipses** lie on no lane: 5,124 (3.5%).

3.3 Stride statistics

Table 2. Stride statistics along the 12 lanes. Partial runs at lane ends are excluded.

Measure	Value
Steps along the 12 lanes	182,640 of 23.5 synodic months; 6,314 of 34.5 synodic months
Runs of 23.5-synodic-month steps between two consecutive 34.5-synodic-month steps	6,302 runs
Run length, in 23.5-synodic-month steps	min 23, max 36, mean 28.9, median 28 (28 steps occurs in 3,822 runs)
Run length, in years	min 43.7, max 68.4, mean 55.0, median 53.2

3.4 Characteristics of the off-stride eclipses

- **Types:** 2,333 partial solar and 2,791 penumbral lunar eclipses. None is total, annular, hybrid, or umbral.
- **Gamma and magnitude:** $|\gamma|$ ranges from 1.2104 to 1.5853 (mean 1.462, median 1.478). Magnitude ranges from below 0.0001 to 0.632 (mean 0.165).
- **Position in the Saros series:** every off-stride eclipse is within 16 eclipses of the first or last eclipse of its series. Saros positions are known exactly for 5,079 of them, all within 15. The other 45 belong to series cut off by the catalog edges, so their positions are estimated; one of these is estimated to be the 16th member of its series.
- **Change over time:** the number per 1,000 years falls across the record, from 228 to 256 in the earliest millennia to 68 in the last complete one, while the total eclipse rate stays near 4,780 per 1,000 years (Figure 1).

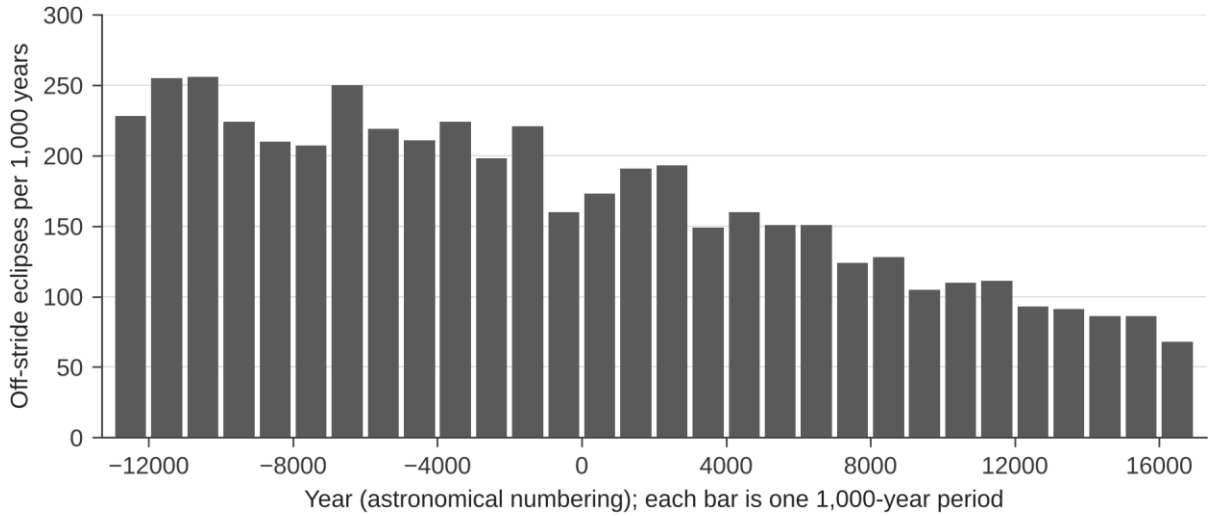


Figure 1. Off-stride eclipses per 1,000 years, from −13000 to +16999. Each bar is one 1,000-year period (e.g. the first bar is −13000 to −12001). The partial periods at the two ends of the catalog are omitted. The values can be recalculated from the off-stride eclipse list in the repository (Section 8).

4. The Shared Interval Set

4.1 How it was found

1. **Intervals measured.** For every off-stride eclipse, the signed interval in cumulative seconds to every other eclipse in the catalog was measured.
2. **Two directions.** The intervals apply in opposite directions depending on where the off-stride eclipse sits in its Saros series: Direction A for eclipses near the start of their Saros series (2,524) and Direction B for eclipses near the end (2,600). Throughout this report, intervals are given as they apply from a Direction B eclipse: the target lies at the starting time plus the interval. From a Direction A eclipse, the target lies at the starting time minus the interval. Each off-stride eclipse matches the full set in exactly one direction with corresponding negative and positive intervals, and that direction corresponds to its Saros position.
3. **When an interval qualifies.** An interval is kept when it lands on an eclipse (on-stride or off-stride) from every off-stride eclipse whose target falls inside the catalog's test range (Appendix A). Where an interval's target would fall before the start or after the end of the catalog, that interval could not be tested from that eclipse and was counted as neither a hit nor a miss. Because orbital timing drifts slowly, an interval is a narrow band rather than one exact number. All of its measured values must fit inside a single window of twice the tolerance. The tolerance is ±246,896 seconds (±2.86 days), half the width of the widest band observed.

4.2 Results

Table 3. Properties of the shared interval set. Reach is stated in the Direction B sign convention.

Measure	Value
Intervals in the shared set	16,377. Every one lands on an eclipse from every off-stride eclipse (0 misses)
Reach	−4,862 to +4,761 years; 8,288 reach into the past and 8,089 into the future, depending on the off-stride eclipse's direction
Whole vs half lunations	8,258 are a whole number of lunations (solar to solar or lunar to lunar); 8,119 are a whole number plus a half (solar to lunar or lunar to solar)
Most common spacings between neighboring intervals	177/178 days (one eclipse season), 162/163, 14/15, 517, 694 (23.5 synodic months), 339/340, and 6,585 days (one Saros)
Band width, median by reach	under 1,000 yr: 1.9 days; 1,000 to 2,000: 2.8; 2,000 to 3,000: 3.8; 3,000 to 4,000: 4.6; over 4,000: 5.1; widest: 5.72 days

4.3 Tests from all Saros-edge eclipses

The shared-interval test was repeated from a much larger population: every eclipse among the first or last N members of every Saros series, on-stride and off-stride alike, for N = 15, 10, 5, 3, and 1. First members were tested in Direction A and last members in Direction B. An interval counts only if it lands on an eclipse from every test eclipse whose target falls inside the test range, with a tolerance of ±3.5 days (302,400 seconds). As in Section 4.1, an interval whose target would fall before the start or after the end of the catalog was not tested from that eclipse (counted as neither a hit nor a miss). To make certain that the first and last N members tested are the true first and last members of their Saros series, eclipses within about 270 years (about 15 Saros periods) of each end of the catalog were removed from the test populations. A series cut off by a catalog edge has an unknown number of members beyond that edge, so its apparent first or last members there cannot be counted exactly. Table 4 summarizes the results.

Table 4. Shared intervals from all Saros-edge eclipses, by window size. Reach is stated in the Direction B sign convention. Percentages are of the intervals with zero misses ("also in" and "not in") and of the 16,377 intervals in the shared set ("share of the shared set").

Measure	First/last 15	First/last 10	First/last 5	First/last 3	First/last 1
Test eclipses	59,498	39,665	19,837	11,905	3,971
Intervals with 0 misses	8,268	10,506	13,038	14,137	15,270
Reach (years)	−3,486 to +3,555	−3,768 to +4,087	−4,428 to +4,296	−4,464 to +4,761	−4,862 to +4,761
Also in the shared set	8,260 (99.9%)	10,488 (99.8%)	12,902 (99.0%)	13,888 (98.2%)	14,846 (97.2%)
Not in the shared set	8	18	136	249	424
Share of the shared set reached	50.4%	64.0%	78.8%	84.8%	90.7%

The results show four regularities:

1. **Nested sets.** Every interval that passes a wider window also passes each narrower one.
2. **Fewer test eclipses, more shared intervals.** As the window narrows from 15 to 1 member, the number of zero-miss intervals nearly doubles (8,268 to 15,270) and the reach widens from about

3,500 to about 4,800 years in each direction. At $N = 1$ the reach equals that of the shared set (−4,862 to +4,761 years).

3. **Close agreement with the shared set.** Between 97.2% and 99.9% of the zero-miss intervals at each window are also in the shared set found from the off-stride eclipses alone. At $N = 1$, the first and last members of every Saros series share 90.7% of the shared set.
4. **Near-complete intervals at the series edges.** At $N = 1$, a further 612 intervals miss only 1, 2, or 3 of the 3,971 test eclipses (276, 204, and 132 intervals respectively). Their 1,080 misses fall on 341 first or last members, all of them partial solar (156) or penumbral lunar (185) eclipses; 144 are off-stride and 197 on-stride. The most frequently missed are the partial solar eclipses of 05/16/(12416) (50 intervals) and 10/31/(9661) (39 intervals).

4.4 Null tests from random Saros-edge populations

To test whether a population of Saros-edge eclipses with the same size and make-up as the off-stride eclipses would share a comparable interval set by chance, the shared-interval test was repeated from random populations. Each population contains 5,124 eclipses, the same number as the off-stride set. They were drawn with a random number generator (NumPy default_rng, with a fixed seed for each test) so that the number of eclipses at each Saros position matches the off-stride population; for example, 20.5% are the first or last member of their series and 16.2% the second or second-to-last. Only eclipses whose position from the start or end of their series is known exactly were eligible, so Saros series cut off by the catalog edges were excluded. Three populations were drawn from all eclipses (Table 5) and three from on-stride eclipses only (Table 6). As in Section 4.3, eclipses counted from the start of their series were tested in Direction A and those counted from the end in Direction B, with the same tolerance (±3.5 days), test range, and treatment of targets outside the catalog.

Table 5. Null tests from random populations drawn from all eclipses, matched to the off-stride population in size and Saros position. Reach is stated in the Direction B sign convention. Percentages are of the intervals with zero misses ("also in") and of the 16,377 intervals in the shared set ("share of the shared set").

Measure	Null test 1	Null test 2	Null test 3
Test eclipses	5,124	5,124	5,124
Off-stride eclipses included	766	756	785
Intervals with 0 misses	9,875	9,663	9,780
Reach (years)	−3,683 to +3,744	−3,609 to +3,764	−3,609 to +3,640
Also in the shared set	9,761 (98.8%)	9,495 (98.3%)	9,626 (98.4%)
Not in the shared set	114	168	154
Share of the shared set reached	59.6%	58.0%	58.8%

Table 6. Null tests from random populations drawn from on-stride eclipses only, matched to the off-stride population in size and Saros position. Reach is stated in the Direction B sign convention. Percentages are of the intervals with zero misses ("also in") and of the 16,377 intervals in the shared set ("share of the shared set").

Measure	On-stride null 1	On-stride null 2	On-stride null 3
Test eclipses	5,124	5,124	5,124
Intervals with 0 misses	10,447	10,507	10,315
Reach (years)	−3,683 to +4,038	−3,750 to +4,020	−3,665 to +3,744
Also in the shared set	9,601 (91.9%)	9,661 (91.9%)	9,448 (91.6%)
Not in the shared set	846	846	867
Share of the shared set reached	58.6%	59.0%	57.7%

The populations drawn from on-stride eclipses only have substantially more zero-miss intervals that are not in the off-stride shared set (846 to 867) than the populations drawn from all eclipses (114 to 168), each of which includes 756 to 785 off-stride eclipses.

Every random population shares far fewer intervals than the off-stride eclipses do: 9,663 to 9,875 zero-miss intervals for populations drawn from all eclipses and 10,315 to 10,507 for on-stride populations, compared with 16,377 for the off-stride eclipses (Section 4.2). At the same ±3.5-day tolerance and ±5,000-year search used for the random populations, the off-stride eclipses still share exactly 16,377 intervals. The null sets also reach less far, about 3,600 to 4,000 years in each direction, compared with −4,862 to +4,761 years for the shared set, and each reaches only 57.7% to 59.6% of the shared set. For populations drawn from all eclipses, 98.3% to 98.8% of the zero-miss intervals are also in the shared set; for on-stride populations the figure is 91.6% to 91.9%, leaving 846 to 867 zero-miss intervals outside the shared set. The high number of shared intervals among the off-stride eclipses (16,377) therefore appears to be unique and significant: other eclipse populations with the same size and the same relation to the start and end of their Saros series share only 9,663 to 10,507.

A random population of 5,124 interior eclipses, none within 15 eclipses of the start or end of its Saros series, was also tested (Table 7). Because an interior eclipse does not clearly face toward either end of its series, its direction was assigned from the data: batches of direction changes were kept only when they improved the result, and the final directions agree with the nearer end of the series for 95.0% of the eclipses. Even so, the interior population shares only 7,473 intervals, fewer than the first/last 15 test (8,268, from a population more than ten times larger; Table 4) and well below the 9,663 to 10,507 of the Saros-edge null populations. Only 2,498 of them (33.4%) are in the off-stride shared set, reaching 15.3% of it, and the reach is shorter, −3,301 to +3,314 years. Together with the rise in shared intervals as the window narrows from 15 members to 1, this indicates that the shared intervals become stronger as eclipses near the start or end of their Saros series.

Table 7. Null test from a random population of interior eclipses, none within 15 eclipses of the start or end of its Saros series (both Saros positions 16 or higher; series cut off by the catalog edges excluded). Directions were assigned from the data (see text). Reach is stated in the Direction B sign convention. Percentages are of the intervals with zero misses ("also in") and of the 16,377 intervals in the shared set ("share of the shared set").

Measure	Interior null 1
Test eclipses	5,124
Off-stride eclipses included	0
Intervals with 0 misses	7,473
Reach (years)	−3,301 to +3,314
Also in the shared set	2,498 (33.4%)
Not in the shared set	4,975
Share of the shared set reached	15.3%

5. Off-Stride Cycles and Long-Term Change

5.1 Definition

An *off-stride cycle* is a group of off-stride eclipses of the same direction. Two consecutive off-stride eclipses of the same direction belong to the same cycle unless more than 150 years separate them; any threshold from 137.6 to 171.8 years gives the same cycles. The record contains 112 cycles, alternating A, B, A, B. Where consecutive cycles overlap in time, their eclipses interleave, and each eclipse belongs to the cycle of its own direction. For example, the Direction B eclipse of 04/28/16961 belongs to cycle 110: it falls 137.6 years after the previous Direction B eclipse (cycle 110) and 171.8 years before the next one (the first eclipse of cycle 112).

Cycle 1 begins with the first off-stride eclipse, 08/28/(13192). Cycle 112 ends with the last, 10/08/17185. Both are limited by the catalog edges.

The *interval from the prior cycle* is the time from the last eclipse of one cycle to the first eclipse of the next.

- **Clean break:** a positive interval, i.e. a gap with no off-stride eclipses.
- **Overlap:** a negative interval. Here the next cycle starts before the previous one ends, and the two directions interleave eclipse by eclipse during the overlap.

5.2 Characteristics

Table 8. Characteristics of the 112 off-stride cycles. The full cycle list is given in Appendix C (Table C1).

Measure	Value
Cycle length (first to last eclipse)	52.7 to 346.9 years; mean 194.1; median 183.4
Eclipses per cycle	12 to 88; mean 45.8; median 41.5
Transitions (111)	98 clean breaks of 2.7 to 156.7 years; 13 overlaps of 2.5 to 60.4 years
Start-to-start interval between same-direction cycles (two cycles apart)	380 to 718 years; mean 548; median 536

The longest overlap, 60.4 years, occurs twice: between cycles 3 and 4 and between cycles 7 and 8. The last transition, between cycles 111 and 112, is a 78.7-year clean break. Cycle 112 is cut off by the catalog's end.

5.3 How cycles interact: an example period, 1000 to 3000

Figure 2 shows every off-stride eclipse between 1000 and 3000. The window contains 384 off-stride eclipses in cycles 45 to 52: 197 Direction A and 187 Direction B. Cycle 45 begins in 935 and cycle 52 ends in 3065, so both are shown only in part.

- Clean breaks in this window range from 15.6 years (cycles 45 to 46) to 120.6 years (cycles 46 to 47).
- One overlap: cycle 50 (Direction B) begins on 05/14/2291, 4.4 years before cycle 49 (Direction A) ends on 09/24/2295.

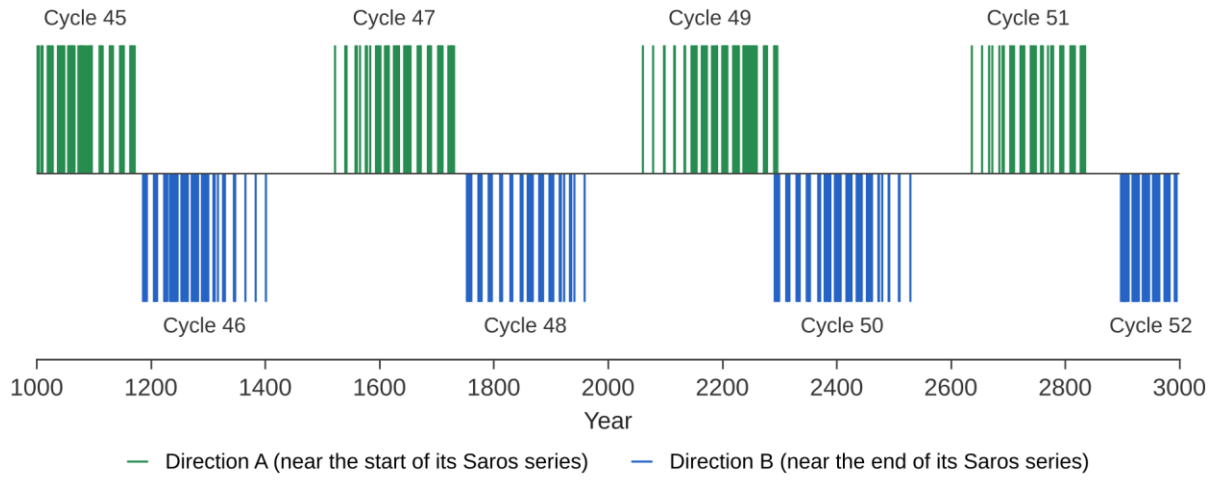


Figure 2. Off-stride eclipses from 1000 to 3000, by direction. Each line is one off-stride eclipse: Direction A (near the start of its Saros series) in green, drawn upward; Direction B (near the end of its Saros series) in blue, drawn downward. Cycles 45 and 52 extend beyond the window.

5.4 Long-term change

Figures 3 and 4 show the eclipse count and length of every cycle across the record.

- **Eclipses per cycle:** fall from 63 to 88 in the earliest cycles to 12 to 20 in the latest (Figure 3).
- **Cycle lengths:** fall from about 250 to 350 years to about 90 to 150 years (Figure 4). A few long, sparse cycles appear near the end of the record (cycle 106: 188.4 years, 15 eclipses; cycle 110: 255.3 years, 15 eclipses).
- **Total eclipse rate:** stays near 4,780 per 1,000 years throughout (Section 2.2).

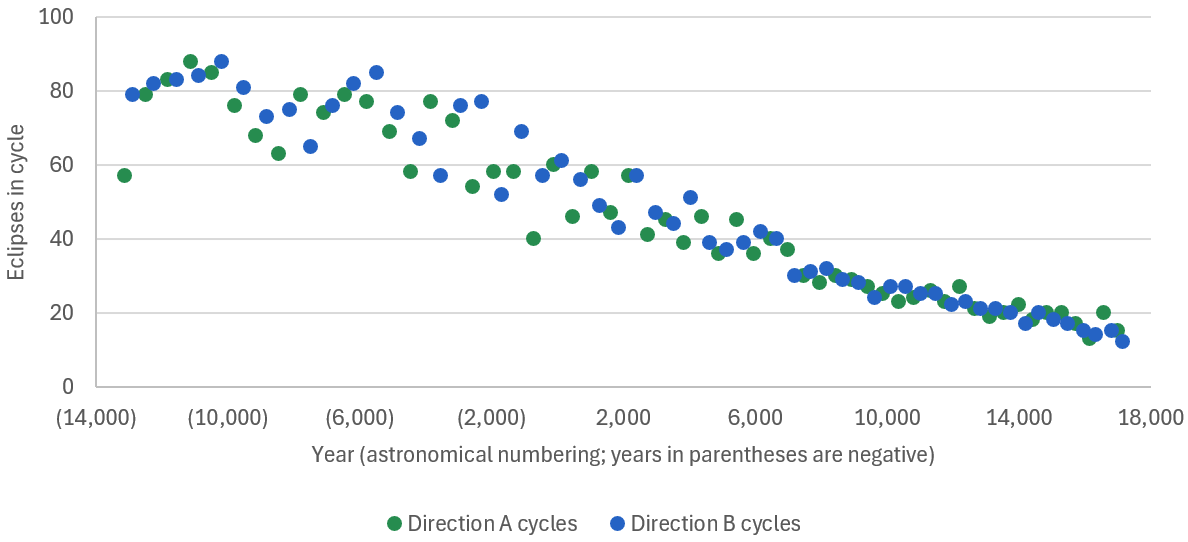


Figure 3. Off-stride eclipses per cycle, plotted at each cycle's midpoint, by direction (Direction A green, Direction B blue).

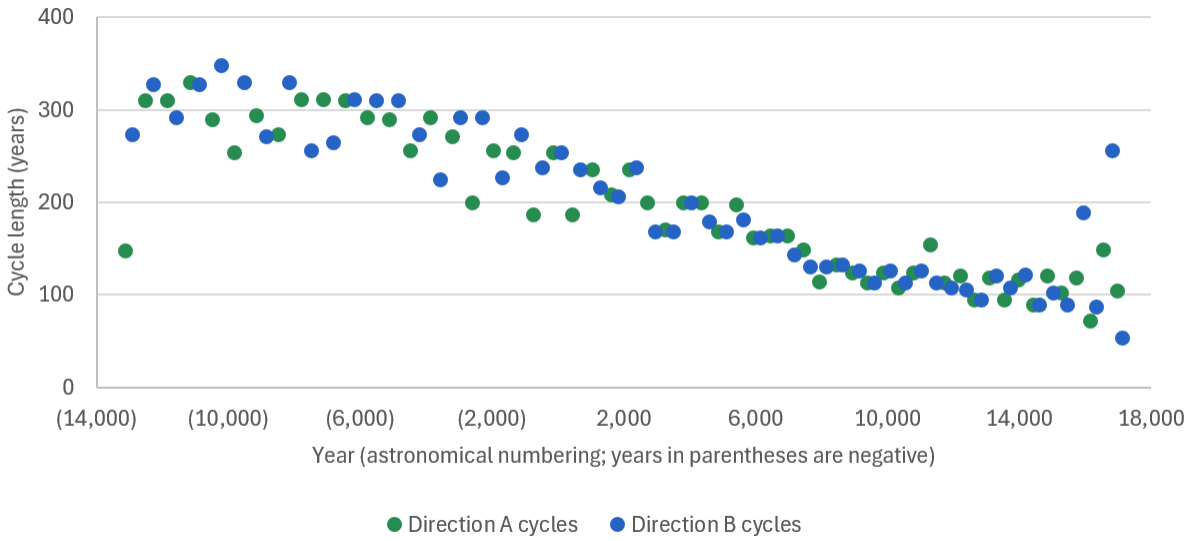


Figure 4. Off-stride cycle length, from the first to the last eclipse of the cycle in years of 365.242 days, plotted at each cycle's midpoint, by direction. The last point (cycle 112) is cut off by the catalog's end.

6. Additional Points That Share the Full Interval Set

Starting points other than the off-stride eclipses were tested against all 16,377 intervals: every on-stride eclipse, and every moment across the catalog's test range. The tolerance was ±3.5 days (302,400 seconds), measured from the middle of each interval's band. Each interval was tested against the known eclipses in the catalog. Where an interval's target would fall before the start or after the end of the catalog, that interval could not be tested from that starting point and was counted as neither a hit nor a miss. Two further groups match the full set; no other starting points do.

6.1 Matching on-stride eclipses: 2,854

- **Type:** all are partial solar or penumbral lunar eclipses, with |γ| from 1.149 to 1.583.
- **Saros position:** all lie within 14 eclipses of the start or end of their Saros series.
- **Direction:** the matching direction follows the same rule as for off-stride eclipses: Direction A near the start of the series, Direction B near the end.

6.2 Matching eclipse near-misses: 2,993

These are moments with no eclipse that nevertheless match all 16,377 intervals.

- **New or full moons:** every one falls at a new or full moon, with a median of 0.23 days from the exact moment.
- **Position relative to a Saros series:** nearly all sit a whole number of Saros periods beyond the edge of an existing Saros series.
  - Direction A points: 1,491 fall before a series' first eclipse; none fall after a series' last.
  - Direction B points: 1,389 fall after a series' last eclipse; none fall before a series' first.
  - 113 (3.8%) cannot be assigned cleanly.

7. Summary

A 30,388-year eclipse catalog computed from the DE441 ephemeris was examined with a fixed 23.5/34.5-synodic-month stride rule. The results are:

1. The stride rule's six forward and six backward lanes reach 140,232 eclipses (96.5%). The 5,124 eclipses they never reach are all partial solar or penumbral lunar eclipses with |γ| between 1.2104 and 1.5853, and all lie within 16 eclipses of the start or end of their Saros series.
2. A single set of 16,377 intervals, reaching from −4,862 to +4,761 years, leads from every off-stride eclipse to another eclipse with no misses. It applies in one direction from eclipses near the start of a Saros series and with the opposite sign from eclipses near the end. Random eclipse populations with the same size and the same relation to the start and end of their Saros series share only 9,663 to 10,507 intervals (Section 4.4), so the high number shared by the off-stride eclipses appears to be unique and significant.
3. In time order, the off-stride eclipses form 112 cycles that alternate between the two directions. Across the record, the number of eclipses per cycle falls from 63–88 in the earliest cycles to 12–20 in the latest, and cycle length falls from about 250–350 years to about 90–150 years, while the total eclipse rate stays constant.
4. The full interval set is also shared by 2,854 on-stride eclipses and by 2,993 new or full moons at which no eclipse occurs. Nearly all of the latter sit a whole number of Saros periods beyond the start or end of an existing Saros series.

The catalog, code, and key data files are published so that each of these results can be recalculated independently. The author welcomes recalculation and testing of the results, and further mathematical or astronomical explanation of the regularities described here.

8. Data, Code, and Recalculation

GitHub repository: https://github.com/mslave231/Eclipse-stride-pattern-with-off-stride-eclipses-sharing-intervals-and-forming-lt-cycles

- **Catalog file:** eclipse_catalog_full_-13198_17189_detailed.csv (engine output, including γ, magnitudes, and Saros positions).
- **Engine and pipeline:** eclipse_catalog.py, the pipeline/ scripts, and scripts for comparison with NASA and Liu.
- **Pattern scripts (pattern/ folder):** lanes.py (lanes and the on-stride/off-stride split), shared_intervals.py (the shared interval search), matching.py (every point in time that matches all shared intervals), cycles.py (off-stride cycles), and null_tests.py (the seven null tests). Each script checks its results against the published data files.
- **Supporting data and detail:** CSV files of the stride lanes, off-stride eclipses, shared intervals, matching points, off-stride cycles, and null tests, with a data dictionary, using the terminology of this report.

Table 9. Key numbers to reproduce.

Quantity	Value
Eclipses	145,356
Lanes	6 forward + 6 backward
On-stride / off-stride	140,232 / 5,124
Direction A / B	2,524 / 2,600
Shared intervals	16,377
Off-stride cycles	112
Matching on-stride eclipses	2,854
Matching eclipse near-misses	2,993

Acknowledgements

- **Yuk Tung Liu:** solar and lunar eclipse catalogs, −3999 to +4000 (https://ytliu.epizy.com/eclipse/). Used as the independent comparison catalog.
- **NASA:** Eclipse Predictions by Fred Espenak (NASA's GSFC).
- **Luca Quaglia and John Tilley:** Saros-Inex panorama of solar eclipses, −11000 to +15000, computed with an integrator built on Aldo Vitagliano's Solex (https://eclipse.gsfc.nasa.gov/SEsaros/SEpanorama.html).
- **JPL Solar System Dynamics group:** the DE441 ephemeris.
- **Open-source software:** Skyfield, NumPy, and openpyxl.

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Appendix A. Parameters and Definitions

Table A1. Parameters used in the analysis.

Parameter	Value
Time axis	Cumulative seconds since −15000 Jan 1 00:00 TD (Julian before 1582 Oct 15, Gregorian after)
Stride lengths	23.5 synodic months, then 34.5 synodic months if no eclipse at 23.5
Stride tolerance	±1.5 days (129,600 s)
Lane seeds	Starting eclipses at the start (forward lanes) and end (backward lanes) of the catalog identified as true lanes by matching missed eclipses to eclipses that match all off-stride shared intervals (Section 3.2)
Shared-interval band	All measured values within one window of 2 × 246,896 s; tolerance ±2.86 days
Full-set match test (Section 6)	±3.5 days (302,400 s) from each band's midpoint
Test range	Targets between cumulative seconds 56,881,000,000 and 1,015,815,000,000
Direction A / B	Off-stride eclipse near the start / end of its Saros series; intervals applied with opposite signs
Interval sign convention	Stated as applied from a Direction B eclipse (target = start + interval); from a Direction A eclipse, target = start − interval
Off-stride cycle	Same-direction off-stride eclipses less than 150 years apart (137.6 to 171.8 years gives the same 112 cycles)
Year length used for cycle statistics	365.242 days
"~"	Saros position estimated because the series is cut off by a catalog edge

Appendix B. Glossary

On-stride eclipse: an eclipse reached by at least one lane.

Off-stride eclipse: an eclipse reached by no lane.

Lane: one of the 6 forward or 6 backward walks produced by the stride rule.

Saros position: an eclipse's place in its Saros series, counted both from the first member (position relative to the start) and from the last member (position relative to the end); position 1 is the first or last member itself.

Inex: the 10,571.95-day interval (358 lunations) linking neighboring Saros series.

Shared interval set: the 16,377 intervals that lead from every off-stride eclipse to another eclipse.

Direction A / Direction B: the two directions in which the shared intervals apply, associated with off-stride eclipses near the start and the end of their Saros series respectively.

Off-stride cycle: a group of off-stride eclipses of the same direction, with no more than 150 years between consecutive members.

Clean break: a gap between consecutive cycles in which no off-stride eclipse occurs.

Overlap: a span where the next cycle begins before the previous one ends.

Matching on-stride eclipse: an on-stride eclipse that matches all 16,377 intervals.

Matching eclipse near-miss: a new or full moon with no eclipse that matches all 16,377 intervals.

Appendix C. Full Cycle Table

Length runs from the first to the last eclipse of the cycle, in years of 365.242 days. The interval from the prior cycle runs from the last eclipse of the previous cycle to the first eclipse of this one; a negative value is an overlap. Dates and times are TD; negative years are shown in parentheses.

Table C1. The 112 off-stride cycles.

Cycle	Dir.	First eclipse (TD)	Last eclipse (TD)	Eclipses	Length (yr)	Interval from prior cycle (yr)
1	A	08/28/(13192) 09:33:43	04/12/(13045) 12:56:45	57	146.6	–
2	B	12/10/(13032) 21:18:05	10/09/(12759) 04:52:16	79	272.8	13.7
3	A	05/15/(12656) 01:50:18	04/04/(12347) 20:13:14	79	308.9	102.6
4	B	11/26/(12408) 14:41:20	10/28/(12081) 06:33:42	82	326.9	−60.4
5	A	06/03/(11978) 03:39:44	04/23/(11669) 08:59:01	83	308.9	102.6
6	B	11/30/(11692) 04:42:58	10/09/(11401) 20:59:38	83	290.9	−22.4
7	A	06/21/(11300) 05:57:16	04/15/(10971) 17:47:56	88	328.8	100.7
8	B	12/07/(11032) 12:02:54	11/08/(10705) 08:42:08	84	326.9	−60.4
9	A	06/02/(10620) 21:51:11	05/18/(10331) 13:25:39	85	289.0	84.6
10	B	12/26/(10354) 02:49:58	10/31/(10007) 08:53:24	88	346.9	−22.4
11	A	05/26/(9922) 22:29:25	04/20/(9669) 05:39:15	76	252.9	84.6
12	B	01/12/(9675) 21:15:06	11/08/(9347) 05:17:24	81	328.8	−6.3
13	A	07/30/(9293) 15:05:06	05/03/(9000) 07:03:14	68	292.8	53.7
14	B	12/11/(8958) 23:01:23	11/16/(8687) 01:15:33	73	270.9	42.6
15	A	07/22/(8595) 16:57:42	05/22/(8322) 03:49:21	63	272.8	91.7
16	B	01/24/(8299) 02:22:44	11/19/(7971) 12:30:36	75	328.8	22.7
17	A	07/30/(7935) 13:53:07	05/14/(7624) 13:24:42	79	310.8	35.7
18	B	01/17/(7601) 10:50:28	11/05/(7347) 16:22:37	65	254.8	22.7
19	A	08/18/(7257) 20:43:10	06/02/(6946) 10:15:04	74	310.8	89.8
20	B	03/02/(6943) 08:37:38	12/24/(6680) 16:55:51	76	263.8	2.7
21	A	07/20/(6595) 06:42:40	06/09/(6286) 22:26:32	79	308.9	84.6
22	B	02/11/(6328) 22:38:58	11/26/(6018) 04:31:13	82	310.8	−42.3
23	A	08/08/(5917) 14:15:26	06/17/(5626) 10:52:18	77	290.9	100.7
24	B	01/25/(5648) 06:12:14	12/14/(5340) 11:47:58	85	308.9	−22.4
25	A	07/21/(5237) 10:10:40	07/05/(4948) 08:49:27	69	289.0	102.6
26	B	02/01/(4988) 19:05:56	12/22/(4680) 11:42:59	74	308.9	−40.4
27	A	08/28/(4588) 07:00:31	06/17/(4333) 16:42:08	58	254.8	91.7
28	B	02/09/(4328) 09:02:08	12/08/(4056) 18:30:33	67	272.8	4.6
29	A	09/09/(3984) 13:36:46	07/20/(3693) 18:35:42	77	290.9	71.8
30	B	01/22/(3648) 17:13:17	01/06/(3424) 03:50:20	57	224.0	44.5
31	A	08/12/(3322) 02:19:31	07/17/(3051) 02:29:46	72	270.9	102.6
32	B	02/23/(3073) 21:27:44	01/02/(2782) 19:46:05	76	290.9	−22.4
33	A	09/25/(2664) 14:28:59	07/18/(2465) 13:32:53	54	198.8	118.7
34	B	02/20/(2431) 05:42:24	12/31/(2141) 12:25:09	77	290.9	33.6
35	A	10/02/(2069) 07:45:56	07/21/(1814) 10:09:49	58	254.8	71.8

Cycle	Dir.	First eclipse (TD)	Last eclipse (TD)	Eclipses	Length (yr)	Interval from prior cycle (yr)
36	B	03/25/(1791) 09:18:44	02/01/(1565) 08:28:33	52	225.9	22.7
37	A	09/07/(1463) 07:24:22	08/01/(1210) 18:52:56	58	252.9	102.6
38	B	03/21/(1214) 19:38:57	01/18/(941) 17:14:00	69	272.8	−4.4
39	A	09/05/(821) 00:55:55	09/03/(635) 00:11:25	40	186.0	120.6
40	B	03/18/(572) 05:23:19	12/24/(336) 17:07:13	57	236.8	62.5
41	A	10/06/(246) 21:21:48	08/31/7 10:36:30	60	252.9	89.8
42	B	03/14/5 19:07:31	02/05/258 22:50:30	61	252.9	−2.5
43	A	09/11/360 22:17:29	09/11/546 03:08:31	46	186.0	102.6
44	B	03/16/591 01:16:58	01/27/826 17:25:37	56	234.9	44.5
45	A	10/14/935 19:25:52	08/28/1170 07:00:24	58	234.9	109.7
46	B	03/22/1186 19:29:33	02/28/1401 15:45:55	49	214.9	15.6
47	A	10/15/1521 23:41:30	08/24/1729 13:48:31	47	207.8	120.6
48	B	04/28/1752 11:34:58	04/04/1958 04:00:11	43	205.9	22.7
49	A	11/08/2060 04:04:11	09/24/2295 04:59:53	57	234.9	102.6
50	B	05/14/2291 04:20:22	02/21/2528 12:36:42	57	236.8	−4.4
51	A	12/16/2635 02:16:25	10/08/2834 21:43:26	41	198.8	107.8
52	B	04/23/2897 02:43:06	04/13/3065 05:10:15	47	168.0	62.5
53	A	12/10/3203 22:14:10	10/23/3373 16:11:26	45	169.9	138.7
54	B	05/08/3436 21:12:37	04/27/3604 09:27:28	44	168.0	62.5
55	A	12/13/3724 18:19:50	10/08/3923 06:55:59	39	198.8	120.6
56	B	05/13/3957 05:49:16	03/05/4156 09:30:01	51	198.8	33.6
57	A	01/08/4282 08:42:50	11/01/4480 08:32:01	46	198.8	125.8
58	B	06/18/4532 07:41:04	05/08/4711 11:16:56	39	178.9	51.6
59	A	12/12/4813 11:41:27	12/01/4981 02:15:04	36	168.0	102.6
60	B	05/11/5046 17:20:05	04/29/5214 23:25:31	37	168.0	64.4
61	A	12/16/5334 08:16:17	11/15/5531 22:24:11	45	196.9	120.6
62	B	06/20/5565 19:16:16	04/03/5746 02:49:52	39	180.8	33.6
63	A	02/06/5872 01:21:17	12/13/6032 19:43:23	36	160.9	125.8
64	B	06/24/6086 04:30:37	05/03/6247 03:02:26	42	160.9	53.5
65	A	03/18/6391 10:51:02	12/18/6553 01:34:43	40	162.8	143.9
66	B	07/24/6587 00:26:53	04/25/6750 15:06:00	40	162.8	33.6
67	A	03/09/6894 23:10:56	12/11/7056 06:51:33	37	162.8	143.9
68	B	08/08/7126 12:50:05	06/04/7269 00:09:47	30	142.8	69.7
69	A	01/20/7390 09:38:21	02/04/7538 11:40:43	30	148.0	120.6
70	B	06/19/7622 16:01:23	06/22/7752 15:51:40	31	130.0	84.4
71	A	03/02/7909 18:27:55	01/17/8023 06:42:11	28	113.9	156.7
72	B	07/08/8105 04:51:42	07/12/8235 07:28:43	32	130.0	82.5
73	A	03/20/8392 10:24:44	02/16/8524 02:55:56	30	131.9	156.7
74	B	07/25/8588 17:32:50	06/23/8720 11:38:29	29	131.9	64.4
75	A	03/01/8877 14:17:42	01/23/9000 01:51:24	29	122.9	156.7
76	B	07/19/9091 19:30:05	05/04/9216 22:20:20	28	124.8	91.5
77	A	03/20/9360 06:21:30	03/12/9472 22:07:39	27	112.0	143.9

Cycle	Dir.	First eclipse (TD)	Last eclipse (TD)	Eclipses	Length (yr)	Interval from prior cycle (yr)
78	B	08/07/9574 09:48:05	07/29/9686 19:22:02	24	112.0	102.4
79	A	03/28/9825 14:00:44	02/19/9948 01:17:19	25	122.9	138.7
80	B	08/14/10039 16:19:06	05/30/10164 22:01:02	27	124.8	91.5
81	A	04/16/10308 06:20:13	01/20/10415 00:51:12	23	106.8	143.9
82	B	08/21/10504 22:51:20	08/14/10616 10:39:01	27	112.0	89.6
83	A	04/13/10755 05:31:53	03/04/10878 16:55:10	24	122.9	138.7
84	B	08/29/10969 05:29:13	06/15/11094 13:08:24	25	124.8	91.5
85	A	04/30/11238 21:56:13	01/26/11392 15:11:46	26	153.7	143.9
86	B	09/06/11434 12:15:11	08/30/11546 02:10:43	25	112.0	42.6
87	A	04/26/11685 21:20:20	04/19/11797 07:36:54	23	112.0	138.7
88	B	09/13/11899 19:11:25	06/19/12006 19:54:30	22	106.8	102.4
89	A	07/22/12155 19:02:37	02/19/12275 07:23:44	27	119.6	149.1
90	B	10/22/12353 05:56:57	09/02/12458 09:36:09	23	104.9	78.7
91	A	05/13/12615 13:21:59	04/24/12709 14:36:00	21	93.9	156.7
92	B	09/18/12811 01:46:07	08/30/12905 09:05:51	21	93.9	102.4
93	A	07/15/13049 17:43:28	03/21/13167 16:59:23	19	117.7	143.9
94	B	10/15/13247 05:44:37	05/14/13367 15:38:48	21	119.6	80.6
95	A	05/06/13509 12:57:24	04/17/13603 14:34:54	20	93.9	142.0
96	B	11/16/13692 13:09:24	08/23/13799 08:14:39	20	106.8	89.6
97	A	07/09/13943 16:48:44	04/20/14059 01:39:34	22	115.8	143.9
98	B	11/14/14139 13:26:00	05/07/14261 15:39:54	17	121.5	80.6
99	A	08/11/14388 04:44:27	05/05/14477 11:21:42	18	88.7	127.3
100	B	10/30/14568 05:41:19	07/25/14657 14:17:58	20	88.7	91.5
101	A	10/03/14804 00:30:49	05/02/14924 11:46:21	20	119.6	147.2
102	B	01/03/15003 10:11:16	07/22/15104 14:09:05	18	101.5	78.7
103	A	11/05/15249 11:49:55	05/25/15351 14:30:11	20	101.5	145.3
104	B	10/12/15444 21:55:12	07/08/15533 05:41:19	17	88.7	93.4
105	A	09/14/15680 15:52:28	05/21/15798 15:02:35	17	117.7	147.2
106	B	12/16/15878 03:15:14	05/05/16067 01:44:42	15	188.4	80.6
107	A	10/19/16125 03:52:33	07/02/16196 01:34:33	13	70.7	58.5
108	B	12/02/16307 19:30:18	10/02/16394 13:57:08	14	86.8	111.4
109	A	11/19/16505 06:32:05	05/30/16654 08:43:09	20	148.5	111.1
110	B	01/13/16706 05:39:03	04/28/16961 03:33:11	15	255.3	51.6
111	A	12/22/16950 17:56:21	06/04/17054 10:44:41	15	103.4	−10.3
112	B	02/04/17133 08:42:21	10/08/17185 00:45:40	12	52.7	78.7